

Enriching Tourist UX via a Location Based AR Treasure Hunt Game

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Abstract—In this research we implemented an Augmented Reality (AR) treasure hunt game, to enhance a visitor's enjoyment and learning while touring a culturally rich city. The city chosen for this study was Valletta, the 2018 European Capital of Culture. Two prototype mobile applications were implemented and deployed to investigate the effectiveness of an AR serious game with in-built navigation tools. A pragmatic mixed-methods research approach was adopted, with data collected, analysed and triangulated using game log data, pre-post test interviews, observations and a survey. It was concluded that AR is useful to provide city information and enriched the tourist's experience. However, the experience needs to be provided by implementing a treasure hunt 'learning through fun' approach, and including GPS navigation crucial to elicit interest and achieve a competitive advantage over traditional city tours.

Index Terms—Augmented Reality, Tourism, Education, Heritage, User Experience, Gamification, Entertainment.

I. INTRODUCTION

Augmented Reality (AR) is considered to be a new medium of creative expression that can enrich the manner in which human beings experience reality [1], [2]. Applications such as SnapChat ¹ and Pokmon GO ² have led to an increase in awareness and everyday use of AR. The overlay of this digital information onto the users' surroundings provides an opportunity to multiple industries and scenarios, including the enhancement of the tourism experience. The tourism industry is one of the main contributions to the gross domestic product in many countries and has a direct impact on the development of a country's economy. Traditional city tours are frequently marred by time constraints, varying visitor interest levels and communication issues. Marketing a destination has become challenging and many researchers in the tourism industry argue that it has become more competitive [3]. Thus, AR can be the new modern technology which will up rise a destination through a mobile-based virtual experience which modifies current consumers' prior, en route, and post-experience [4]. AR applications were found to create meaningful, enjoyable and enhanced tourism experiences [5].

The hypothesis of this study is that a location driven AR treasure hunt game enriches a user's experience in terms of learning and entertainment, during a heritage tour. Given this hypothesis three main questions were formulated:

- 1) How does an AR enhanced city tour compare with a traditional walk-around?
- 2) What UX features are required to elicit interest and what are the potential inhibitors intimidating visitors when using the AR application?
- 3) How can AR be used to provide an improved experience of Maltese Heritage and will individuals learn from this process?

Our study starts with a review of literature, structured in relevant hyper-themes. Tourism AR projects were investigated to inform the methodology of this study and UX requirements for the development of two AR prototype applications. Given the nature of the study a justification for mixed-methods approach is provided. Following the on-site experiment, data collected from pre-post test interviews, application log data and survey were analyzed both separately and triangulated together. In the final section, a summary of conclusions, limitations and recommendations for further research are provided.

II. BACKGROUND

A. AR in Tourism

The use of modern technology is crucial for many destination-marketing organizations. Many museums and historical sites have already started to make use of media, such as movie clips and photography to enhance visitor experience [6]. However, this approach many times lacks the potential of modern interactive technology [6]. AR provides significant benefits for many industries due to its nature of mixed environment, or computer enhancement of real world setting [7]–[9].

As [10] outlines, the proper use of AR for in the tourist heritage sector still requires investigation to enhance the tourist experience. However, if implemented well, AR has the potential of becoming a mainstream technological tool in the tourism industry in the future due to its indoor and outdoor capabilities which will enhance the tourist experience, learning and fun.

B. AR for Learning

Educators have recognized that Virtual Reality (VR) has hidden benefits that help students in education, namely history, science and mathematics [11]. Moreover, [12], found that the feelings of presence that VR may induce can assist the learning process. Whilst VR is more concerned with immersion [12],

¹<https://www.snapchat.com/>

²<https://www.pokemongo.com/en-us/>

AR's potential for learning can be exploited in museums, heritage areas, and tourist attractions, using a basic smartphone device. In line with Gardner's theory of multiple intelligences, which suggests that "traditional psychometric views of intelligence are too limited and that all people have different kinds of intelligences" [13], multimedia elements can aid learning via a multi-sensory approach. By embedding different media like images, video, text and 3D models within an AR application, different learning styles are catered for. For example, some people prefer visual learning over auditory learning [13].

Due to its potential, as stated by [14], "AR is changing the face of education and creating new opportunities for improvement in quality of teaching and learning and trying to reflect the user's information needs". [15]'s systematic literature review about AR found that AR-based systems were more effective in improving learners' motivation and satisfaction than traditional ones, especially for situated, inquiry based, and self-regulated learning. However, the review found that improved applications still lack adeptness, with them being too much emphasizing on entertainment and generally being limited by simple visuals [15]. On the other hand, [16], found that usability or ease-of-use appear to be the biggest challenge in some applications but then again, most studies concluded that AR enhances enjoyment, motivation and interaction for learners.

C. AR in Entertainment

In the entertainment sector, AR is mainly used for interactive games which combine the real world and virtual world together. With the aid of AR, the user can play with 3D characters imposed on real time backgrounds. In addition, with the use of mobile phones, some organizations introduced AR in their magazine which requires the reader to point the camera to a marker and animated characters appear on the mobile screen [17]. In this scenario AR is used to give life to text stories through further multi-sensory reader immersion. In 2016, the release of the 'Pokémon GO' AR game, took the world by a storm. The game quickly became a worldwide phenomenon, mainly by exploiting AR in a time of significant developments in mobile devices. This marked a sudden leap in mass awareness and popularity of AR. The scope of the game was to catch animated Pokémon appearing in the real-world, which were visible through the smartphone's camera.

D. Related AR Projects

Several authors list multiple examples of using AR in tourism, namely, augmented walks, where tourists are placed within the real environment, but with the possibility to view additional artificial information in the form of 3D reconstructions of monuments, either via digital screens, or head-mounted displays [18]–[22]. In 2009, the region of Tuscany created the first AR smartphone application which was created to guide tourists around Tuscany by adding a layer of guidance, content and entertainment to physical locations seen through the smartphone's camera view [23].

The city of Vienna already provides a tourist guide application which guides users to different locations via navigation, as well as is able to provide location-based information on nearby places, which can be selected at will and share this information via social media [24]. Another example of a destination that aims to enhance the overall tourist experience through AR is Dublin. The initiative to utilize AR to strengthen and promote the city of Dublin as an 'innovative city' started in 2012 as a cooperation between the Manchester Metropolitan University, the Dublin Institute of Technology and the City Council of Dublin. [3] stated, "This application will provide a platform to superimpose tourism relevant information, reconstruct and revive stories of the past, assisting the tourist in creating an emotional experience of the intangible product". This application will not only be of benefit to tourists but also to citizens who want to learn about relevant information about the history of a 'An Post Museum' of the city. This application contains historical text, pictures and videos through AR which aid users to gain information about a site to create awareness and enhance the tourists' experience in the context of Dublin's historical heritage. For the users who had little or no idea of the concept of AR, a map highlighting AR-enabled contents in and around the 'An Post Museum'. Moreover, because of the success of AR in the Museum, Dublin decided to also create an AR application to help with the outside city which increased steadily the tourists' experience of learning about new cities [25], strengthen and promote the city of Dublin, [26].

E. UX requirements for Mobile/Handheld AR applications

According to [25], identification of UX requirements for AR applications is limited. A reoccurring theme is simplicity and design of the user interface [18]–[21]. Good performance of the application with less reaction and loading times [18]–[21]. Therefore, the size of the application needs to be controlled to facilitate the installation process. Safety and security issues are a priority [18]–[21]. Application should be accessible at any time and available on different platforms [18]–[20]. Given the high probability of city tour time-constraints, the application needs to be efficient to attract utilization [18]–[21]. A search and browse button feature makes the AR application more user friendly to access context-aware relevant information [22]. Finally, proper use of good user interface designs are a must in an application [3]. These UX requirements should aim to increase the overall tourist's experience in retrieving personalized multimedia information by means of a user-friendly, multi-language interface.

III. METHODOLOGY

A. Research Methods

Given that this study needs to connect quantitative data with an interpretative investigation of human behaviour, a pragmatic mixed methods approach was adopted [27]. This approach has also been previously applied in tourism and AR sectors, where a quantitative approach was ideal to examine topics such as user requirements in a mobile computing context [3] and a qualitative approach, to examine the impact of cultural



Fig. 1. AR 3D Model of original Royal Opera House (destroyed in WWII)

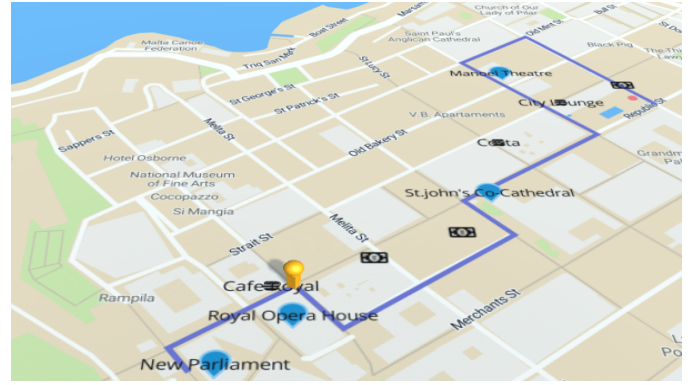


Fig. 3. UX2 GPS Live Map

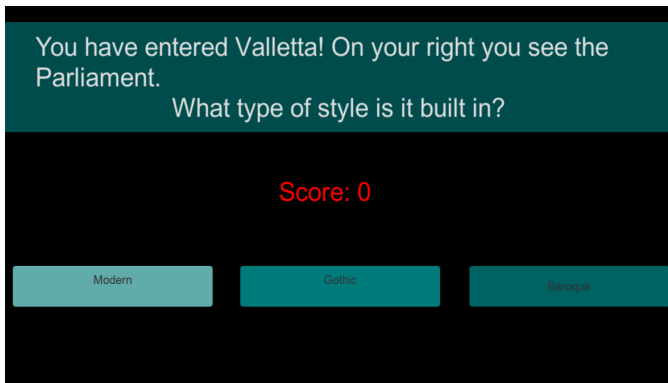


Fig. 2. UX1 Game Interface

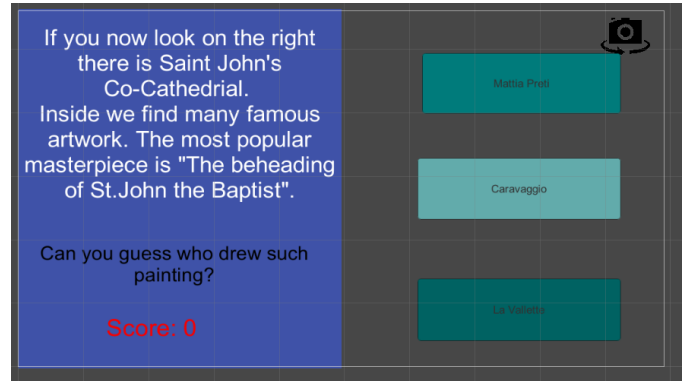


Fig. 4. UX2 Game Interface

differences on behavioral intentions for a more accurate use of AR [28].

B. Prototype Applications

Two prototypes were developed with similar on-location marker-based AR features (Figure 1), but with a different UX design. This decision was made to test the effectiveness of location-based navigation in a modified UI to better complement AR features. UX1 only had a static map and buttons at the bottom (Figure 2), with a location driven UI implemented in UX2 (Figure 3, Figure 4). In both applications, the user simply points the smartphone to a target using the camera. Information about Valletta is given through text, images, video and 3D models, with the user answering a question about the point of interest. Game points are added or subtracted according to the user responses. The applications were developed with Unity³ as game engine, Vuforia for AR⁴, MapBox⁵ for location-based game features and Maya for 3D modelling⁶.

³<https://unity.com/>

⁴<https://developer.vuforia.com>

⁵<https://www.mapbox.com/>

⁶<https://www.autodesk.com/products/maya/overview>

C. Survey Design

The survey questionnaire was divided in two sections. The first section gathered age/gender demographic data and presented generic questions about AR. The second section provided a demo of the game, after-which a series of questions where provided to capture the respondents views on the prototype. These questions seek to investigate whether AR functions improve the visitor's experience.

D. Pre-Post Test Interviews and Data Logger

Six participants, who had experienced a traditional city walk-around, were invited for a pre-test interview, following which they made use of the AR applications. After the experience a post-test interview was conducted. The pre-test interviews provided control data with the post-test interviews supplying the experimental data. While roaming around Valletta to locate the markers, location readings and behavior data were recorded to a log file. This methodology was employed to triangulate interview transcripts with tracking data, thus enabling a cross-examination of results with multiple data sources.

IV. RESULTS AND DISCUSSION

Using a mixed method approach, survey, pre-post test interviews and data log files saved on user's devices were analysed first separately and consequently triangulated. Congruent with

our hypothesis, the main objective of this evaluation was to investigate if and how individuals prefer to use AR during a walk-around. In addition, the effectiveness of different UX designs in terms of the overall tour experience, learning and entertainment.

A. Survey Analysis and Discussion

Table I outlines survey demographics which attracted 174 responses from participants, which were invited through a social media campaign. However, all participants had to satisfy the pre-requisite of having experienced a traditional Valletta city walk-around. Prior to experiencing the AR demo 56.9% of

TABLE I
SURVEY DEMOGRAPHICS

Age	Female	Male	Total
11-14	2		2
15-20	22	16	38
21-30	21	20	41
31-50	58	14	72
51-59	12	2	14
60+	4	3	7
Total	119	55	174

respondents answered that they are familiar with AR. Table II shows that male respondents are more aware about AR than females.

TABLE II
DO YOU KNOW WHAT AR IS? (PRIOR TO AR DEMO EXPERIENCE)

Gender	Yes	No
Female	47.06%	52.94%
Male	78.18%	21.82%
Total	56.90%	43.10%

1) *Learning through AR*: Both the question “An application with AR functions allows specific useful information to be communicated better.” (Q7), and the question “An application with AR functionality helps people combine real world with virtual world information” (Q10), address a similar theme as there is no statistically significant difference between responses (Table III). The results show that participants agree that AR facilitates information acquisition.

TABLE III
AR USEFUL TO ACCESS INFORMATION (POST AR DEMO EXPERIENCE)

	Mean	N	Std. Deviation	S.E Mean
Q7	1.96	174	.98	.07
Q10	1.97	174	1.01	.08
Pair 1	Q7 - Q10	t	df	SIG. (2-tailed)
		-.09	173	.927

Table IV summarizes how the majority of the respondents use online applications/websites to access information about a place, replacing paper-based brochures or guides which were still popular until a few years ago.

In a subsequent question, we explored if an AR application “makes it more fun to learn about a place”. The results showcased in Figure 5, demonstrate that participants overwhelmingly agree to this statement.

TABLE IV
PREFERRED MEDIUM TO ACCESS INFORMATION ABOUT A PLACE

	Maybe	No	Yes
Online Application/ Online Websites	36	1	112
Tourist Brochure	5	0	9
Tourist Guide	3	1	7
Total	44	2	128

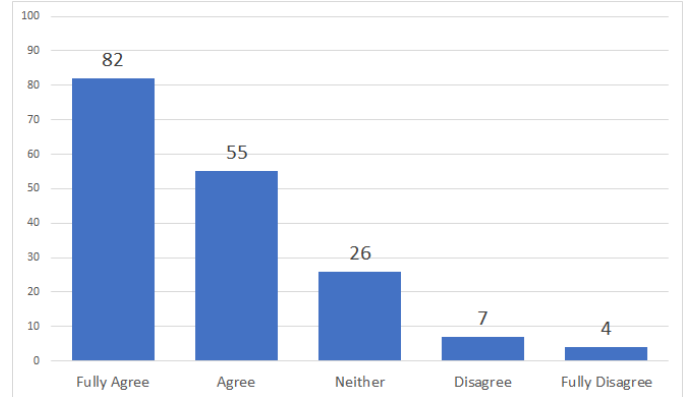


Fig. 5. AR functions are fun to learn about a place

2) *Readiness to use the AR application*: Following a demonstration of the AR application, 41% answered positively in readiness to use the AR application, while 52% were indecisive. Only 7% answered negatively. On further analysis employing a correlation test, respondents not ready to use the AR application, also view AR as difficult to use, signifying that proficiency is a factor that impacts readiness.

B. Pre-Post Test Interviews

The six participants were interviewed prior to using the application. After choosing a device, the participants toured the city via the treasure-hunt game. Following the walk-around, participants recounted their experience. Data tracking each user's behaviour was collected in real-time within the application and automatically exported, upon completion of the tour. Pre-post test interviews and device data logs were analysed first separately and then triangulated. For this purpose, the six participants were randomly assigned codes from RP1 to RP6. RP1-RP4 chose UX1 (Live Map version) and RP5-RP6 (static map).

C. Tracking Data Analysis and Triangulation with Interviews

Tracking data analysis was conducted with all participants. Figure 6 depicts an example of our analysis using RP1's GPS track report, which is consistent with the technique employed with other participants. This data is generated from different co-ordinates during logging, which then formed a track of the pathway the user went through to reach the markers to finish the game.

RP1's total time to complete the tour was of 13 minutes (4 minutes in motion & 9 minutes idle), with a total distance of 1.03km and average speed of 4.93km/h. Figures 7 and 8 show

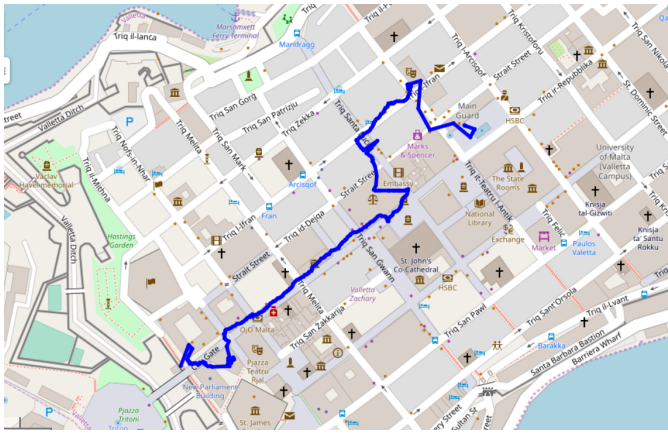


Fig. 6. RP1 GPX trace route

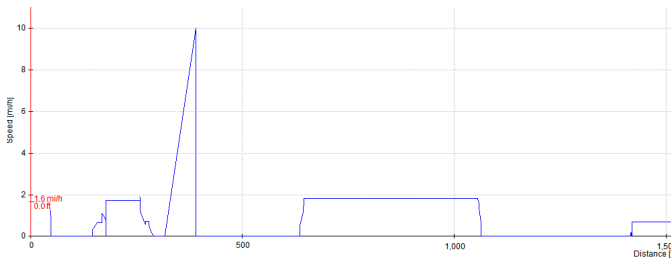


Fig. 7. RP1 - Speed vs Distance Graph (Part One)

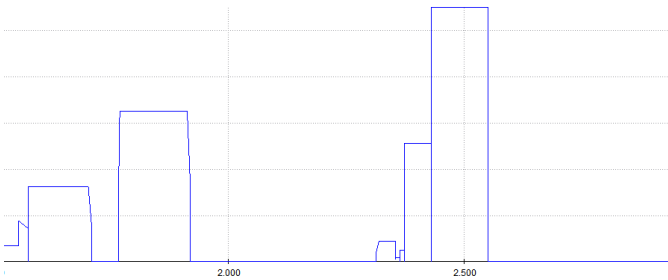


Fig. 8. RP1 - Speed vs Distance Graph (Part Two)

the speed (y-axis) along the route distance(x-axis). RP1 spent most of the time in idle when stopping to locate the marker and answer the question next to the point of interest. Given that participant was familiar with the streets of Valletta, RP1 moved with ease between venues. The location where RP1 took the most time was next to St. John's Co-Cathedral. The place where RP1 was walking fast was from Manoel Theatre to Sette Giugno Monument. A similar analysis was performed for all six participants.

All participants, with the exception of RP4 and RP5, never made use of AR. There was no noticeable benefit of having prior experience of AR as time taken by RP1-RP5 was very similar (UX2: 13-16 minutes). However, while participants navigating with the live map (UX2) took an average of 14 minutes to complete the treasure-hunt, the two participants using the static map (UX1) took an average of 25 minutes. These results confirmed the need of GPS navigation tools

within the AR treasure-hunt game.

D. UX Design

There were mixed preferences regarding buttons positioning, with RP1, RP2 preferring bottom positioning and RP3 and RP4 at the side. All participants found the hints and icons throughout the game to be useful. In pre-test interviews UX1 users thought it would be more challenging and fun to locate the markers the traditional way, rather than using the live map and hints. Following the experience, while enjoying the extra challenge, participants suggested some hints would have been useful. Tracking data also confirmed the difficulty of participants navigating without hints. The two media which were deemed most appealing and useful for learning, were the 3D models and video elements.

With regards to UX improvements, RP1, being a young person, suggested adding a reward system using animations. The other five respondents appreciated the minimalistic and green complementary colours in the game. RP5, being the oldest person from the group at age 65+ suggested to have bigger buttons for easier and quicker access, also adjust the font size according to the liking of the user and also include a voice-over the text.

Adding different languages for a better experience was another remark, as also suggested by [19] and [20]. Following the experience, RP3 recommended extending the use of AR to museum tours.

E. Analysis with other studies

[10], argued that AR is of particular interest to the tourism industry, in the way it is capable to reveal information on the surroundings by putting relevant data into a virtual space. Moreover, [29] concluded that this phenomenon is interesting to experiment with in heritage sites. In our survey, we found that an overwhelming number of respondents agree to the usefulness of AR to discover a heritage site.

[24] concludes that the majority of the users appreciate the flexibility provided in the system build, i.e. the ability to use the system as a tour guide, a map or a guidebook. This finding was also confirmed in our survey, with participants finding AR helpful to learn. [28] assert that AR promotes a good atmosphere at a new destination. Similarly, in our study AR was viewed to provide a more enjoyable tour experience.

The live map feature was much appreciated by participants. This result is corroborated in [24]'s study, where all respondents expressed their positive opinion on the location-aware navigation. Finally, [14], stated that UX is crucial for an application to allow users to learn about heritage. UX was an important factor which was noticed by the respondents with positive feedback during the interview conducted.

V. CONCLUSION

Our study is in agreement with existing research that AR has a great potential for tourism. Location driven UI and game elements are two main themes that emerged from our study. Through a mixed-methods approach, we can confidently

conclude that AR does indeed help users to learn in a fun and interactive way. Whilst the initial hypothesis has been accepted, this study also reveals that AR in tourism applications is greatly enhanced through:

- 1) The inclusion of location-based features and Live-maps.
- 2) A UX Design which is easy to learn and use during a fast city tour.
- 3) Gamification.
- 4) Knowledge rich media to provide insights about a place and assist learning.

The inclusion of marker-less AR technology is also suggested. [10] argues that technology needs to be constantly implemented in tourism destinations to stay competitive and attractive for tourists. Although, we managed to create a solution to enhance the tourist's experience using basic equipment and popular tools, a challenge lies in scaling the solution to cover wider areas. Issues like cost-effectiveness, time and equipment can be addressed with the creation of a framework that enables developers to produce content-rich AR related to tourism.

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